Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Bioreactor-Based Cultivation of Functionally Differentiated Muscle Tissue Constructs for Cultivated Meat**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio/Medical		Bio/Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes / Bio-equipment	Bio-engineering Technologies	Industrial Biotechnology	Functional food ingredients
Concept of Proposed Technology	The proposed technology aims to establish a **next-generation manufacturing platform for the large-scale production of homogeneous, functionally differentiated muscle tissue constructs** intended for cultivated meat applications. By integrating **tissue engineering principles with scalable bioprocess design**, the project seeks to standardize the structural, functional, and mechanical parameters required to reproducibly generate food-grade muscle tissue constructs across diverse bioreactor environments. The technology focuses on bridging the gap between biologically valuable three-dimensional cell aggregates and **true muscle tissue–like constructs** that exhibit organized myofiber structures, mechanical integrity, and food-functional properties such as texture and cooking performance. The resulting platform is intended to serve as a **globally reproducible and industrially scalable bioprocess standard** for differentiated muscle tissue production, forming a critical foundation for structured cultivated meat products.			

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Rationale for Joint R&D Project	The Korean SME partners currently possess advanced capabilities in **primary bovine muscle stem cell isolation, edible scaffold development, serum-free media formulation, and large-scale 3D cell cultivation** within bioreactor systems up to industrial scale. These technologies enable efficient production of three-dimensional muscle and fat cell aggregates, which are essential for cultivated meat manufacturing. However, the aggregates currently achievable do not yet meet the requirements of **functionally differentiated muscle tissue**. Key limitations include the absence of organized myofiber alignment, insufficient mechanical and functional properties, and lack of food-functional characteristics such as meat-like texture and sensory quality. Furthermore, inducing full muscle differentiation at industrial scale remains technically and economically challenging due to low differentiation efficiency, high media consumption, and bioprocess configurations not optimized for mass differentiation. Overcoming these bottlenecks requires **deep mechanistic understanding of tissue formation, cell–material interactions, and mechanobiology**, as well as expertise in translating tissue engineering concepts into scalable biomanufacturing systems. These capabilities cannot be sufficiently developed by a single industrial entity alone. International joint R&D with a research institute specializing in **multi-scale tissue engineering, biomaterials design, mechanobiology, and translational bioprocess development** is therefore essential. This collaboration enables the integration of fundamental scientific insight with applied industrial scale-up, ensuring scientific rigor, global credibility, and practical implementation.
Key Words	1. Cultivated Meat (Cell-cultured Meat) 2. Functional Muscle tissue Biomanufacturing 3. Large-scale Tissue Bioprocessing 4. Scalable Bioreactor Cultivation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (7)
R&D Objectives	**Overall Objective** To develop and validate a **scalable bioreactor-based process** capable of producing homogeneous, functionally differentiated, food-grade muscle tissue constructs suitable for industrial cultivated meat production. **Specific Objectives** • Identify optimal structural, material, and biophysical parameters for stable muscle tissue construct formation under large-scale bioreactor conditions • Establish quantitative **Critical Quality Attributes (CQAs)** defining biological differentiation, structural integrity, and food-functional performance • Develop scalable bioprocess strategies that prevent unwanted construct aggregation or fusion during mass cultivation			

	• Validate construct formation, maturation, and food-functional properties across scale transitions from laboratory to pilot and industrial volumes
R&D Target Specifications	The project targets the following performance criteria: **Structural and Material Targets** • Identification of edible material systems supporting reproducible 3D construct formation • Determination of construct architectures that maintain stability and minimize construct–construct fusion • Establishment of size ranges compatible with nutrient diffusion, differentiation, and shear tolerance • Consistent construct morphology with defined independence and geometric uniformity **Functional Differentiation Targets** • Quantifiable muscle differentiation metrics, including fusion index, alignment index, and sarcomere formation • Maintenance of differentiation and viability across scale transitions (5 L → 50 L → 500 L) • Stability of construct properties under bioreactor hydrodynamic conditions **Bioprocess and Scale-Up Targets** • Definition of bioreactor operating windows supporting large-scale construct maturation • Demonstration of reproducible production at mid- and large-scale volumes • Establishment of bioprocess KPIs such as construct yield, CQA compliance rate, and metabolic performance **Food-Functional Quality Targets** • Definition and validation of texture-related parameters (hardness, cohesiveness, chewiness, elasticity) • Evaluation of water-holding capacity, thermal stability, and cooking loss • Correlation of food-functional properties with construct architecture and differentiation status
R&D Plan	**Year 1 (2026): Material and Construct Architecture Screening** • Screening of edible biomaterials and construct-generation strategies • Comparison of construct architectures and evaluation of fusion tendencies • Investigation of differentiation cues including ECM composition, biochemical factors, and mechanical stimuli • Establishment of preliminary CQAs for biological, structural, and food-functional performance **Year 2 (2027): Mid-Scale Process Development (5 L → 50 L)** • Formation and maturation trials in mid-scale bioreactors • Optimization of agitation, seeding, feeding, and environmental control parameters • Refinement of CQAs based on scale-dependent outcomes • Development of hydrodynamic and mass-transfer models linking process conditions to construct behavior

	Year 3 (2028): Large-Scale Production and System Integration (500 L) • Execution of repeated 500 L production runs • Validation of construct differentiation, stability, and food-functional performance • Finalization of CQA framework and acceptance criteria • Integration into a modular, container-based production system and generation of process documentation

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will focus on **fundamental scientific validation and design principles**, including: • Investigation of cell–cell and cell–material interactions governing 3D muscle tissue formation • Evaluation of edible material systems and construct architectures for differentiation stability • Identification of biochemical and mechanical cues driving muscle maturation • Development of analytical methods and CQAs related to differentiation, morphology, and fusion behavior • Provision of experimental data and technical guidance to support scale-up and bioprocess design
Expected R&R of Korean SME Partner	The Korean SME partner will lead **applied development, scale-up, and process integration**, including: • Provision of primary muscle stem cells, edible biomaterials, and serum-free media • Implementation of construct-generation strategies within proprietary bioreactor systems • Execution of cultivation trials at 5 L, 50 L, and 500 L scales • Optimization of mixing, flow, feeding, and seeding strategies to ensure construct stability • Validation of CQAs at scale, including differentiation and food-functional metrics • Preparation of safety, process, and quality documentation for future regulatory submissions

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Scientific and Technological Impact** • Establishment of a validated, scalable platform for producing functionally differentiated muscle tissue constructs • Generation of internationally competitive scientific publications and intellectual property • Definition of standardized CQAs and operational windows for large-scale muscle tissue bioprocessing **Regulatory and Industrial Impact** • Completion of data packages supporting cultivated meat regulatory submissions • Advancement of structured cultivated meat toward commercial readiness

	• Strengthening of national capabilities in animal-cell bioprocessing and next-generation protein manufacturing **Broader Impact** • Acceleration of the transition from prototype-level cultivated meat to reproducible, food-grade products • Enhancement of global competitiveness of Korean SMEs in cultivated meat and biomanufacturing • Contribution to long-term transformation of food production technologies

6. References

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